

Power-Based Normalization of Team Performance in Mixed Indoor Rowing Relays

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We normalize the results of a mixed-gender indoor-rowing relay, the Råda Rowing Challenge 2026, in which ten five-person teams of different gender composition shared one ergometer each for 30 minutes. Each team is scored by the fraction σ of reference power that its rowers sustained, where the reference is set by world indoor-rowing records. Because relay members row one after another, their *distances* add, not their powers; the team reference matching the measured distance is therefore the power mean with exponent 1/3 of the individual reference powers, and the score reduces to $\sigma = (D/D^{\text{ref}})^3$, independent of the monitor's power-speed constant. Comparing the power at the team's mean speed with the arithmetic mean of reference powers, the intuitive alternative, is biased against mixed teams by up to 1.3 % and is enough to swap the top two places. Expressing the score in power rather than speed does not change the order, but it reports what the athletes actually produce: a 7 % speed deficit is a 20 % power deficit. The ranking is instead sensitive to the choice of reference distance, which we quantify. With 2000 m records as reference, the two leading teams finish in a dead heat, 0.5 m apart.

I. INTRODUCTION

The Råda Rowing Challenge (RRC) is an indoor rowing competition organized by Mölnåls Roddklubb [1], held annually at RådaRum in Mölnlycke, Sweden. The 2026 edition took place on 7 March 2026 and included individual events as well as a 30-minute mixed relay [2]. In the relay, each team of five rowers shares *one* Concept2 RowErg, taking turns on the machine; the team's result is the total distance shown by the monitor after 30 min.

Ten teams entered: Chalmers Roddklubb (two teams), Jönköpings Roddsällskap (two teams), Halmstads Roddklubb, Mölnåls Roddklubb (two teams), Göteborgs Roddklubb, Råda Crew, and CTH-GU. Compositions ranged from all-female (Göteborgs Roddklubb) to four men and one woman (Chalmers Roddklubb – Team 2). Because men and women differ systematically in ergometer output, raw distance favours teams with more men.

We develop a composition-aware score in four steps: the ergometer's power-speed relation (Sec. II), reference performances and the gender gap (Sec. III), a physical model of the relay that fixes how individual references combine (Sec. IV), and the results with a sensitivity analysis (Sec. V).

II. THE ERGOMETER POWER FORMULA

The Concept2 monitor derives a speed v from the angular deceleration of the air-braked flywheel; v is the speed of a virtual boat, not the handle speed. It converts speed to power

with the fixed relation [3]

$$P = cv^3, \quad c = 2.8 \text{ W s}^3 \text{ m}^{-3} = 2.8 \text{ kg m}^{-1}. \quad (1)$$

The cubic law mirrors a boat moving against hydrodynamic drag $F \propto v^2$, for which the propulsive power is $P = Fv \propto v^3$; c is a conversion constant defined by Concept2 rather than a measured property of an individual machine. For a distance d rowed in time t , the average speed is $v = d/t$; in terms of the 500-metre split $t_{500} = 500 \text{ m}/v$,

$$P = c \left(\frac{500 \text{ m}}{t_{500}} \right)^3. \quad (2)$$

For example, $t_{500} = 120 \text{ s}$ (2:00) gives 202.5 W, and the men's 2000 m world record (334.7 s, $t_{500} = 83.7 \text{ s}$) gives 597 W. Figure 1 shows the strong nonlinearity.

Two properties matter below. First, the displayed power is computed from speed; it is not measured at the handle, and force-sensor studies find it differs from the mechanical power delivered by the rower [6]. Our method uses only distances and times, so it is unaffected. Second, c drops out of every score we define (Sec. IV), so its precise value is irrelevant for the ranking.

III. REFERENCE PERFORMANCE

III.1. Indoor rowing records

Table 1 lists reference performances on the Concept2 RowErg. The 500 m, 2000 m and 5000 m entries are world records from

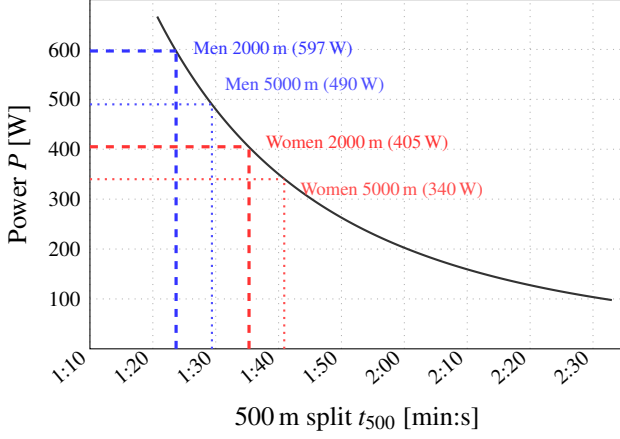


Figure 1: Ergometer power versus 500m split, Eq. (2). Dashed: 2000 m world records (reference used in this study). Dotted: 5000 m world records.

the Concept2 records database [4]; the 1000 m entries are event-winning times from the 2026 World Rowing Virtual Indoor Championships (WRVIC 2026) [5] and are not necessarily records.

Table 1: Reference performances (Concept2 RowErg, open category) and the corresponding power, Eq. (2).

Dist.	Gender	Time [min:s]	P [W]	Athlete (year)
500 m	Men	1:09.8	1029	P. Clapp (2024)
500 m	Women	1:24.5	580	O. Buryak (2017)
1000 m	Men	2:38.0	710	A. Panizza (2026)
1000 m	Women	3:09.4	412	A. Ader (2026)
2000 m	Men	5:34.7	597	O. Zeidler (2026)
2000 m	Women	6:21.1	405	B. Mooney (2021)
5000 m	Men	14:53.9	490	T. George (2022)
5000 m	Women	16:49.4	340	G. Rowe (2026)

III.2. The gender gap in speed and in power

At 2000 m the men's record is faster by a factor $v_m/v_f = 381.1/334.7 = 1.139$, i.e. 14 % in speed, but by $P_m/P_f = (v_m/v_f)^3 = 1.48$ in power. The cubic law turns a moderate speed advantage into a large difference in produced power. The ratio also depends on distance (Fig. 2a): $P_m/P_f = 1.77$, 1.72, 1.48 and 1.44 at 500 m, 1000 m, 2000 m and 5000 m. The men's advantage is largest in short, predominantly anaerobic efforts. The 1000 m values are race results rather than records and should be read with that caveat; the trend is carried by the three record distances.

As shown in Sec. IV, a normalized *ranking* depends on the references only through this ratio: multiplying both reference powers by a common factor rescales every score but leaves the order unchanged. Absolute record levels are therefore irrelevant for the ranking; the choice of reference distance is not.

III.3. Choice of reference distance

Each relay member rows about 6 min in total, which matches a 2000 m effort in duration, and covers about 1.7 km, close to 2000 m in distance. The work is, however, split into short bouts: teams typically swap about every 30 s, so each rower performs roughly a dozen short, hard bouts with recovery in between. Short bouts with recovery typically allow a higher power than one continuous 2000 m effort, and draw more on the anaerobic capacity that dominates the shorter events, where the gender gap is larger. We therefore expect the appropriate ratio to lie at or above the 2000 m value. There is no unique correct reference distance. We adopt the 2000 m records as the primary reference, the standard benchmark in rowing, and report the ranking for all four distances in Sec. V.2.

IV. NORMALIZATION

IV.1. Relay model: distances add

Team k has n rowers, n_m men and n_f women. Rower i rows for a total time τ_i at average speed v_i , with $\sum_i \tau_i = T = 1800$ s. Because the rowers take turns on one machine, the monitor accumulates their distances:

$$D_k = \sum_{i=1}^n v_i \tau_i. \quad (3)$$

Powers do not add in a relay; only one rower produces power at any time. We assume equal shares, $\tau_i = T/n$; then $D_k/T = \bar{v}_k$ is the mean of the rowers' speeds.

IV.2. Uniform-effort hypothesis and the power score

Let $P_i \in \{P_m, P_f\}$ be the reference power of rower i and $v_i^{\text{ref}} = (P_i/c)^{1/3}$ the corresponding speed. The fair question is: what fraction σ_k of their reference power did the rowers of team k sustain? If every rower holds the same fraction, $P = \sigma_k P_i$, then by Eq. (1) each rows at $v_i = \sigma_k^{1/3} v_i^{\text{ref}}$ and Eq. (3) gives

$$D_k = \sigma_k^{1/3} D_k^{\text{ref}}, \quad D_k^{\text{ref}} = \frac{T}{n} \sum_{i=1}^n v_i^{\text{ref}}, \quad (4)$$

where D_k^{ref} is the distance a composition-matched team rowing exactly at reference would cover. Solving for σ_k yields the *power score*

$$\sigma_k = \left(\frac{D_k}{D_k^{\text{ref}}} \right)^3 = \frac{P_k^{\text{team}}}{P_k^{\text{ref}}} \quad (5)$$

with the power at the team's mean speed and its reference

$$P_k^{\text{team}} = c \left(\frac{D_k}{T} \right)^3, \quad (6)$$

$$P_k^{\text{ref}} = c (\bar{v}_k^{\text{ref}})^3 = \left[\frac{1}{n} \sum_{i=1}^n P_i^{1/3} \right]^3. \quad (7)$$

P_k^{ref} is the power mean with exponent 1/3 of the individual reference powers: it equals P_f for an all-female and P_m for an all-male team and increases monotonically with n_m .

Units are consistent throughout: D_k and D_k^{ref} are in metres, σ_k is dimensionless, and P_k^{team} and P_k^{ref} are in watts. Writing the references as distances d rowed in times t_i , so that $v_i^{\text{ref}} = d/t_i$, gives $\sigma_k = [D_k / (\frac{T}{n} \sum_i d/t_i)]^3$. Only distances and times enter; the constant c cancels exactly.

The *speed score* $s_k = D_k/D_k^{\text{ref}}$ is the same information on a different scale, $\sigma_k = s_k^3$. Both give identical rankings.

IV.3. Why score in power

The ranking does not depend on whether the score is expressed in power or in speed, so the case for power is not that it reorders teams. It rests on three points.

(i) *Power is the athlete's output.* Power is the rate at which the rower does work on the machine, and the reference records are power levels that athletes sustained. Under the uniform-effort hypothesis every rower of team k produced $\sigma_k P_i$, so $\sigma_k = 0.8$ means that the rowers sustained 80 % of their reference power. The speed score compresses this: to first order $1 - \sigma_k \approx 3(1 - s_k)$, and still by a factor 2–2.5 for the deficits in this field. A team 7 % slower than reference produced 20 % less power. Speed understates how far teams are from the reference, and how different they are from each other.

(ii) *Power makes the combination rule explicit.* Under the hypothesis, the rowers' mean *produced* power is $\sigma_k \langle P \rangle_k$ with $\langle P \rangle_k = (n_m P_m + n_f P_f)/n$, so σ_k is also the ratio of produced to reference mean power. A relay monitor, however, does not record the mean produced power; it records D_k , i.e. the mean *speed*. The power belonging to that observable is $P_k^{\text{team}} = c \bar{v}_k^3$, the power at the team's mean speed, and its matching reference is the power at the reference team's mean speed, P_k^{ref} of Eq. (7). In a crew boat the situation is reversed: the rowers pull *simultaneously*, the observable boat speed is set by the sum of their powers, and the arithmetic mean of seat powers is the matching reference. Which mean is correct is decided by what the measurement records: summed power for crews, summed distance for relays.

(iii) *Mixing the two is biased.* Dividing P_k^{team} , the power at the team's mean speed, by the arithmetic mean $\langle P \rangle_k$ compares two different quantities. By the power-mean inequality $P_k^{\text{team}} \leq \langle P \rangle_k$, with equality only for single-gender teams, so a mixed team rowing exactly at reference would score below one. The bias grows with the gender gap: at most 1.3 % at 2000 m and 2.8 % at 500 m, largest for 2M/3F (Fig. 2b). This is small, but not negligible here: with the mixed comparison the order of the top two teams reverses (CTH-GU ahead of Chalmers 1).

IV.4. Assumptions

(a) Equal shares of time per rower. Individual rowing times were not recorded; a team that gave its fastest rowers more time would gain relative to the reference. (b) Uniform effort: all rowers of a team at the same fraction of their reference. If instead every rower held the same fraction of reference *speed*, the score would be s_k , and $\sigma_k = s_k^3$ again gives the same ranking. (c) Changeovers: the flywheel coasts while rowers swap, and the monitor keeps counting, so part of T is not rowed by anyone.

The effect depends on the number and speed of changeovers. It is similar for all teams but not modelled; it is contained in D_k . (d) All teams row on Concept2 RowErgs with the same monitor conversion; c cancels in any case. (e) Reference powers are computed from the average speed of the record efforts, $v = d/t$. Records are not rowed at perfectly constant pace, but only these speeds enter the score.

V. RESULTS

V.1. Normalized ranking

With the 2000 m records, $P_m = 597$ W and $P_f = 405$ W. Table 2 and Fig. 3a give the result.

The mean team distance was (8.60 ± 0.76) km (mean $\pm$ sample standard deviation, coefficient of variation 8.8 %). The mean power score was 0.65: on average, the rowers sustained about two thirds of world-record power for their composition, at a mean speed score of 0.858. Scores span 0.34–0.80. All are below one, as expected for a record-based reference.

Dead heat for first place. Chalmers 1 ($\sigma = 0.8033$) and CTH-GU ($\sigma = 0.8031$) are separated by 1.2×10^{-4} . CTH-GU would have needed only 0.5 m more, out of 9.5 km, to draw level. This is below the monitor's 1 m resolution, so the two teams share first place.

Composition matters. Göteborgs Roddklubb, the only all-female team, covered the seventh-longest distance but ranks third. Chalmers Roddklubb – Team 2 (4M/1F) covered the third-longest distance but ranks seventh, because four men set a reference of 555 W.

Scale. Chalmers 1 averaged 1:37.1 per 500 m against a reference pace of 1:30.3 for its composition: a 7 % speed deficit, but a 20 % power deficit, point (i) above.

V.2. Sensitivity to the reference distance

Table 3 and Fig. 3b repeat the analysis for every reference distance. Positions 8–10 are unaffected, and positions 4–7 change by at most two places. First place, however, depends on the reference. With the 500 m records ($P_m/P_f = 1.77$) the all-female Göteborgs Roddklubb wins. With 1000 m or 2000 m, Chalmers Roddklubb – Team 1 wins. With 5000 m ($P_m/P_f = 1.44$), CTH-GU wins. Given how close the top three are, organizers should fix and announce the reference before the event. For an interval-style relay the physiologically most appropriate reference is arguably shorter than 2000 m (Sec. III.3).

VI. CONCLUSION AND OUTLOOK

We have normalized a mixed-gender ergometer relay by the fraction σ of reference power sustained by each team. Because relay members row in sequence, their distances add, and the team reference is the power mean with exponent 1/3 of the individual reference powers. The resulting score, $\sigma = (D/D^{\text{ref}})^3$, needs only distances and times. The arithmetic mean of reference powers, correct for rowers pulling together in one boat, is biased against mixed relay teams when compared with the power at the team's mean speed.

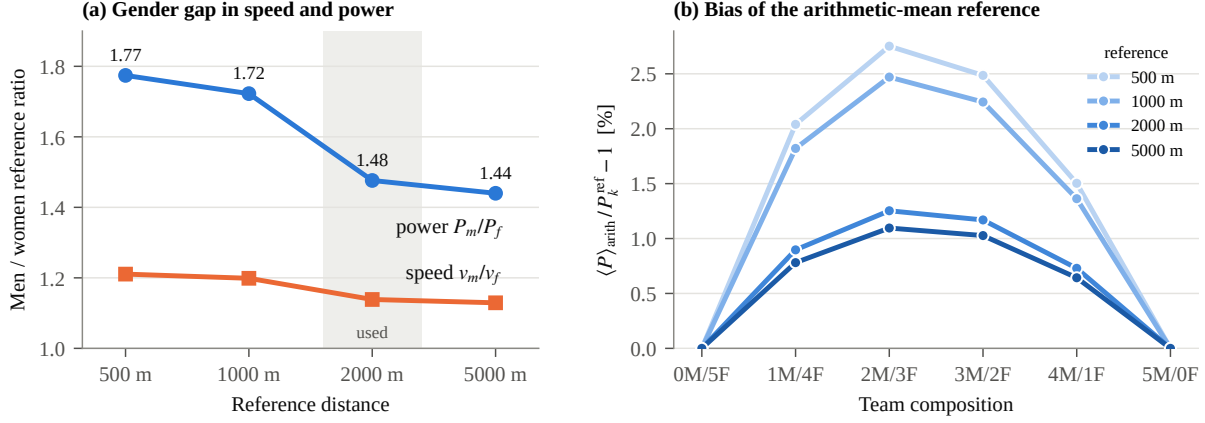


Figure 2: (a) Ratio of men’s to women’s reference speed (squares) and power (circles) versus reference distance, Table 1. The power ratio is the cube of the speed ratio. (b) Relative overestimate of the team reference by the arithmetic mean of reference powers compared with the correct power mean P_k^{ref} , Eq. (7), for all compositions of a five-person team. Single-gender teams are unaffected; mixed teams are penalized, more so for shorter reference distances.

Table 2: RRC 2026 relay results with the 2000 m world records as reference, ranked by power score. D_k : team distance; D_k^{ref} : distance of a composition-matched reference team, Eq. (4); $s_k = D_k / D_k^{\text{ref}}$: speed score; $P_k^{\text{team}}, P_k^{\text{ref}}$: Eqs. (6) and (7); $\sigma_k = s_k^3 = P_k^{\text{team}} / P_k^{\text{ref}}$: power score; last column: rank by raw distance.

Rank	Team	Lane	Comp.	D_k [km]	D_k^{ref} [km]	s_k	P_k^{team} [W]	P_k^{ref} [W]	σ_k	Raw
1	Chalmers Roddklubb – Team 1	8	2M/3F	9.268	9.970	0.9296	382	476	0.8033	2
2	CTH-GU	10	3M/2F	9.511	10.232	0.9295	413	514	0.8031	1
3	Göteborgs Roddklubb	7	0M/5F	8.575	9.446	0.9078	303	405	0.7480	7
4	Jönköpings Rodds. – Team 2	5	2M/3F	9.021	9.970	0.9048	352	476	0.7407	4
5	Mölnö Roddklubb – Team 1	4	2M/3F	8.744	9.970	0.8770	321	476	0.6746	5
6	Mölnö Roddklubb – Team 2	6	1M/4F	8.439	9.708	0.8693	289	439	0.6568	8
7	Chalmers Roddklubb – Team 2	1	4M/1F	9.095	10.494	0.8667	361	555	0.6510	3
8	Jönköpings Rodds. – Team 1	2	3M/2F	8.660	10.232	0.8464	312	514	0.6063	6
9	Råda Crew	9	3M/2F	7.733	10.232	0.7558	222	514	0.4317	9
10	Halmstads Roddklubb	3	2M/3F	6.956	9.970	0.6977	162	476	0.3396	10

Table 3: Rank of each team as a function of the reference distance (Table 1). Teams are ordered by their rank with the 2000 m reference.

Team	Comp.	500 m	1000 m	2000 m	5000 m
Chalmers 1	2M/3F	2	1	1	2
CTH-GU	3M/2F	3	3	2	1
Göteborgs RK	0M/5F	1	2	3	4
Jönköping 2	2M/3F	4	4	4	3
Mölnö 1	2M/3F	6	6	5	5
Mölnö 2	1M/4F	5	5	6	7
Chalmers 2	4M/1F	7	7	7	6
Jönköping 1	3M/2F	8	8	8	8
Råda Crew	3M/2F	9	9	9	9
Halmstads RK	2M/3F	10	10	10	10

Expressing the score in power rather than speed does not change the order, but reports performance on the scale the athletes actually produce, where a few percent in speed is two to three times that in output. What does change the order is the men-to-women reference ratio, set by the choice of reference distance. For RRC 2026 the leading positions are sensitive to

it, and with the 2000 m records the top two teams are in a dead heat.

The method is implemented in the open-source Python package `ergrace` [8], which also generates every number, table and figure in this report. It extends directly to age- and weight-graded references, including the masters handicap categories of World Rowing [7], by replacing P_m and P_f with category-specific values in Eq. (7).

DATA AVAILABILITY

The race data, normalization code, figure scripts and the source of this report are openly available in the `ergrace` repository [8].

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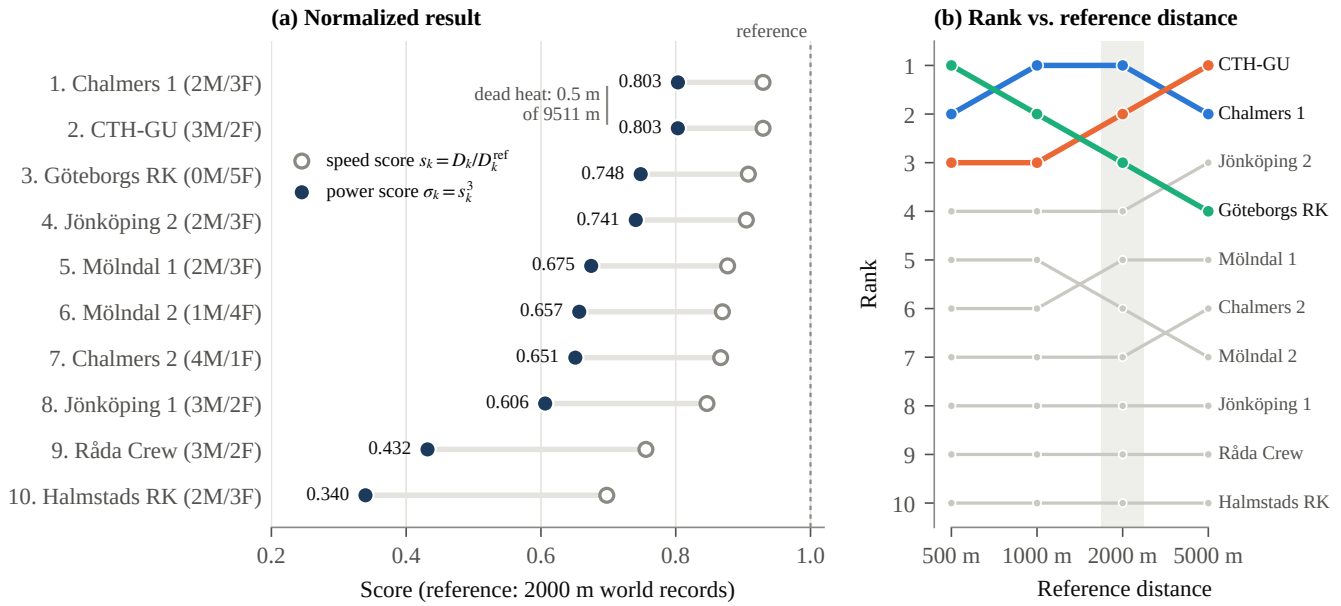


Figure 3: (a) Normalized result with the 2000 m reference: power score σ_k (filled) and speed score s_k (open), connected per team. The power score shows the distance to the reference on the scale of produced power. The top two teams are separated by 0.5 m. (b) Rank of each team for the four reference distances of Table 1; the shaded column is the reference used in (a). First place changes between the top three teams; positions 8–10 do not change.

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AUTHOR CONTRIBUTIONS

LJS wrote the paper and developed the power normalization. JL assisted with data acquisition and analysis. OG acquired the data and supervised the project.

REFERENCES

- [1] Mölndals Roddklubb, <https://www.mrk.nu/>, accessed March 2026.
- [2] Mölndals Roddklubb, “Råda Rowing Challenge – 7 mars 2026,” <https://www.mrk.nu/rada-rowing-challenge-7-mars-2026/>, accessed March 2026.
- [3] Concept2, “RowErg Performance Monitor – pace/power calculator,” <https://www.concept2.com/indoor-rowers/training/calculators>, accessed March 2026.
- [4] Concept2, “World Records,” <https://www.concept2.com/indoor-rowers/racing/records/world>, accessed March 2026.
- [5] World Rowing, “Records fall and champions crowned during 2026 World Rowing Virtual Indoor Championships, presented by Concept2,” <https://worldrowing.com/2026/02/28/records-fall-and-champions-crowned-during-2026-world-rowing-virtual-indoor-championships-presented-by-concept2/>, 28 February 2026.
- [6] S. Boyas, A. Nordez, C. Cornu, and A. Guével, “Power responses of a rowing ergometer: mechanical sensors vs.

Concept2® measurement system,” *Int. J. Sports Med.* **27**, 830–833 (2006), doi:10.1055/s-2006-923774.

- [7] World Rowing, “Rowing Events and Boat Classes,” <http://worldrowing.com/rowing/events-boat-classes/>, accessed March 2026.
- [8] L. J. Splitthoff, *ergrace*: ERG power normalizer for rowing relay competitions, <https://github.com/lukassplitthoff/ergrace-normalizer>, 2026.